

Navneet

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BTECH - CSE

V Semester

Alliance University

1st part

```
clc;  
clear all;  
close all;
```

2nd part

```
d=[1,2;3,4]
```

```
d = 2x2  
    1    2  
    3    4
```

```
a=sin(d)
```

```
a = 2x2  
    0.8415    0.9093  
    0.1411   -0.7568
```

```
sin(90)
```

```
ans =  
0.8940
```

```
A=10;  
t=[1,2,3,4,5,6,7,8,9,10]
```

```
t = 1x10  
    1    2    3    4    5    6    7    8    9   10
```

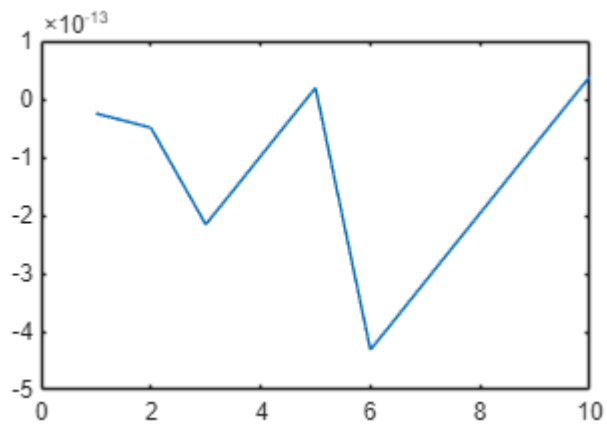
```
f=10
```

```
f =  
10
```

```
ms=A*sin(2*pi*f*t)
```

```
ms = 1x10  
10-12 x  
   -0.0245   -0.0490   -0.2156   -0.0980    0.0196   -0.4312   -0.3136   -0.1959 ...
```

```
plot(t,ms)
```



EXERCISE

create 4X4 matrix and store in a variable x_MAT

```
format long
format short
A=[1 10 ; 3 4]
```

```
A = 2×2
    1    10
    3     4
```

```
max(A)
```

```
ans = 1×2
     3    10
```

```
x_MAT=[1 2 3 4;5 6 7 8;1 2 3 4; 1 2 3 4]
```

```
x_MAT = 4×4
     1     2     3     4
     5     6     7     8
     1     2     3     4
     1     2     3     4
```

```
length(A)
```

```
ans =
     2
```

```
z= ones(2,4+1)
```

```
z = 2×5
     1     1     1     1     1
     1     1     1     1     1
```

```
a_sum=A+10
```

```
a_sum = 2×2
    11    20
    13    14
```

```
a_sub=A-10
```

```
a_sub = 2x2
    -9    0
    -7   -6
```

```
a_mul=A*10
```

```
a_mul = 2x2
    10   100
    30   40
```

```
a_div=A/10
```

```
a_div = 2x2
    0.1000    1.0000
    0.3000    0.4000
```

```
p=A.*A
```

```
p = 2x2
    1    100
    9    16
```

```
a=A^16
```

```
a = 2x2
1014 x
    1.4753    3.5305
    1.0591    2.5345
```

```
n=[2+3i;4+2i]
```

```
n = 2x1 complex
    2.0000 + 3.0000i
    4.0000 + 2.0000i
```

```
B = 0:10:100
```

```
B = 1x11
    0    10    20    30    40    50    60    70    80    90   100
```

Text and Characters

```
t="Hello World"
```

```
t =
"Hello World"
```

```
q="Something ""Quoted"" And Something Else"
```

```
q =
"Something "Quoted" And Something Else"
```

```
whos t
```

Name	Size	Bytes	Class	Attributes
t	1x1	182	string	

```
f=71;
c=(f-32)/1.8;
```

```
temptext= "Temp is"+ c + "C"
```

```
temptext =  
"Temp is21.6667C"
```

3rd part

```
A=["a","bb","ccc";"dddd","eeee","ffffff"]
```

```
A = 2x3 string  
"a"      "bb"      "ccc"  
"dddd"   "eeeeee"  "ffffff"
```

```
strlength(A)
```

```
ans = 2x3  
    1    2    3  
    4    5    6
```

```
seq="123"
```

```
seq =  
"123"
```

```
whos seq
```

Name	Size	Bytes	Class	Attributes
seq	1x1	166	string	

```
abc='Navneet'
```

```
abc =  
'Navneet'
```

```
whos abc
```

Name	Size	Bytes	Class	Attributes
abc	1x7	14	char	

Functions

```
A=[1 2 4];  
max(A)
```

```
ans =  
4
```

```
min(A)
```

```
ans =  
1
```

```
B=[3 4 5]
```

```
B = 1x3  
    3    4    5
```

```
union(A,B)
```

```
ans = 1x5  
      1      2      3      4      5
```

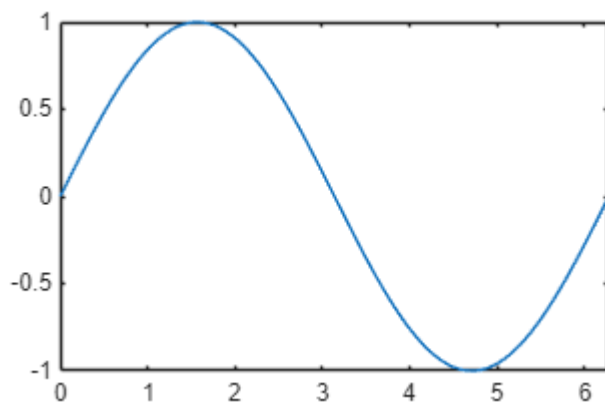
```
disp(abc)
```

Navneet

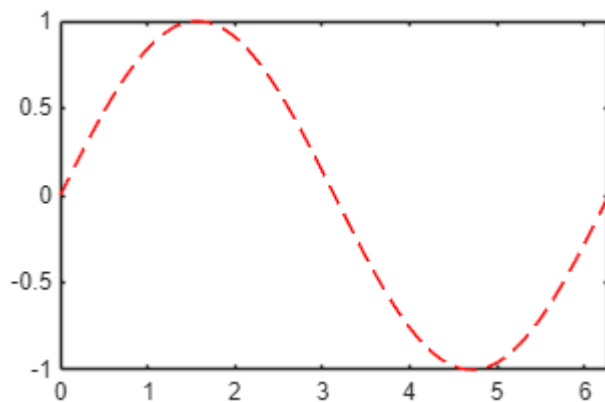
```
x=linspace(0,2*pi)
```

```
x = 1x100  
      0      0.0635      0.1269      0.1904      0.2539      0.3173      0.3808      0.4443 ...
```

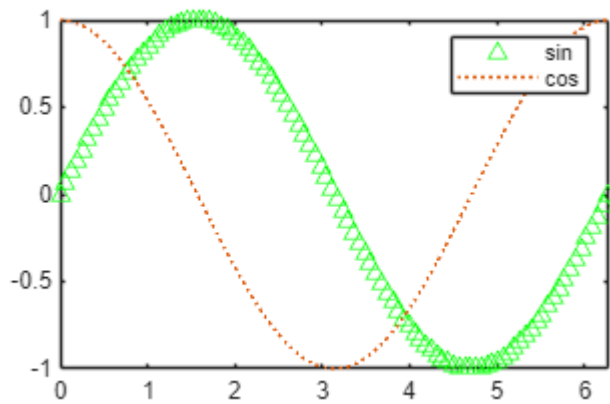
```
y=sin(x);  
plot(x,y)
```



```
plot(x,y,"r--")
```



```
plot(x,y,"g^")  
hold on  
  
y2 = cos(x);  
plot(x,y2,":")  
legend("sin","cos")  
  
hold off
```

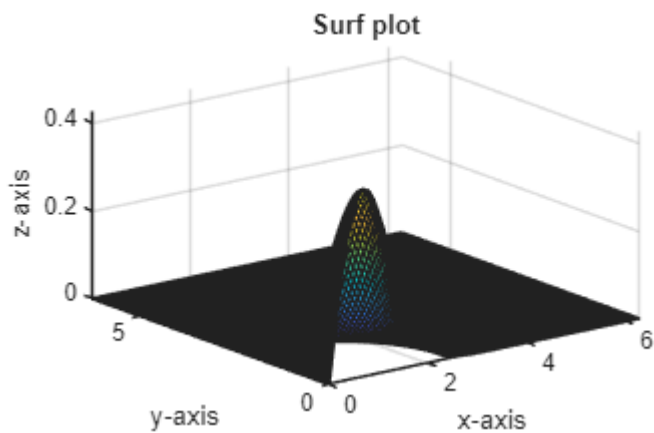


4th part

```
abcd=linspace(-2,2,20);
y=x';
z=x.*exp(-x.^2 - y.^2);
surf(x,y,z)
xlabel("x-axis");
ylabel("y-axis");
zlabel("z-axis");
title("Surf plot");
```

5th part

```
t=tilayout(2,2)
```



```
t =
TiledChartLayout with properties:
```

```
TileArrangement: 'fixed'
GridSize: [2 2]
Padding: 'loose'
TileSpacing: 'loose'
```

Show all properties

```
title(t,"Trignometric functions")
```

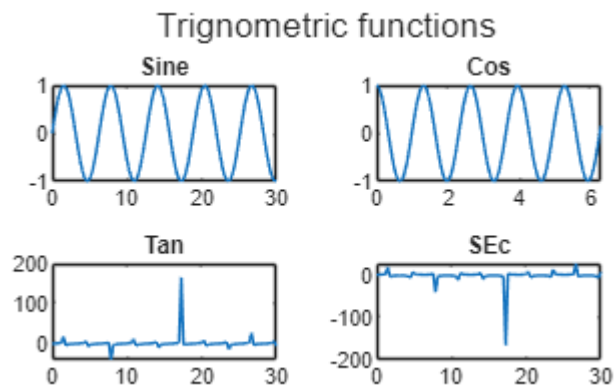
```
x=linspace(0,30);
```

```
nexttile
plot(x,sin(x))
title("Sine")
```

```
nexttile
plot(y,cos(x))
title("Cos")
```

```
nexttile
plot(x,tan(x))
title("Tan")
```

```
nexttile
plot(x,sec(x))
title("SEc")
```



6th part

```
subplot(2,1,1);
x=linspace(0,10);
y1 = sin(x)
```

```
y1 = 1x100
      0      0.1008      0.2006      0.2984      0.3931      0.4839      0.5696      0.6496 ...
```

```
plot(x,y1)
```

```
subplot(2,1,2);
y2=sin(5*x);
plot(x,y2)
```

```
x=linspace(0,2*pi)
```

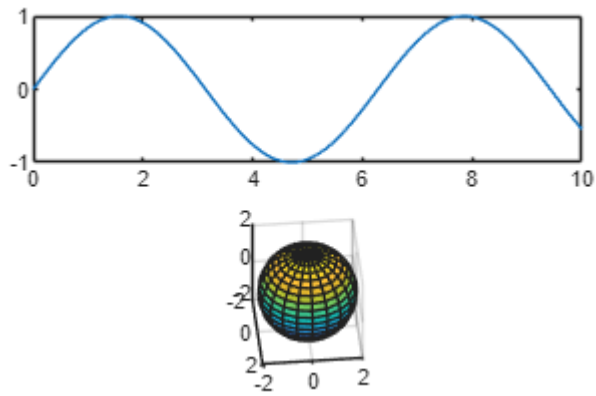
```
x = 1x100
      0      0.0635      0.1269      0.1904      0.2539      0.3173      0.3808      0.4443 ...
```

```
y=sin(x);
stem(x,y)
```

```
bar(x,y)
```

7th part

```
[x,y,z]=sphere;  
r=2;  
surf(x*r,y*r,z*r)  
axis equal
```



```
A=4*pi*r^2;  
v=(4/3)*pi*r^3;
```